# AI-Assisted Customer Insights Pipeline

Ryan Friedman
Customer Review Analytics Case Study

## Problem

Customer reviews contain valuable but unstructured feedback. This project builds an AI-assisted pipeline to identify satisfaction drivers, complaint patterns, and business recommendations from Amazon review data.

## Method

Analyzed 500,000+ Amazon reviews, using a 20,000-review modeling sample split between positive and negative feedback.

- Cleaned raw review text
- Applied TF-IDF vectorization
- Used KMeans clustering
- Used AI-assisted theme labeling

## Key Insights

**1. Product experience drives satisfaction.** Taste, texture, and flavor consistency are major satisfaction drivers.
**2. Logistics create risk.** Shipping delays, damaged products, and incorrect orders create negative sentiment even when product quality is acceptable.
**3. Category-specific drivers matter.** Pet products depend on animal acceptance, while dietary products depend on ingredient trust.
**4. Expectation mismatch drives complaints.** Many negative reviews come from gaps between listing claims and actual experience.

## Business Impact

The analysis shows that product quality alone does not explain customer satisfaction. Customers also react strongly to shipping reliability, product-listing accuracy, category-specific expectations, and perceived value. These findings are useful for product managers, marketers, and operations teams deciding where to improve customer experience.

## Analyst Takeaway

The strongest business opportunity is not simply identifying negative reviews, but understanding which categories create preventable dissatisfaction. The results suggest that companies should prioritize issues that combine high review volume with high negative sentiment, especially logistics, taste consistency, pet acceptance, and dietary-product expectations.

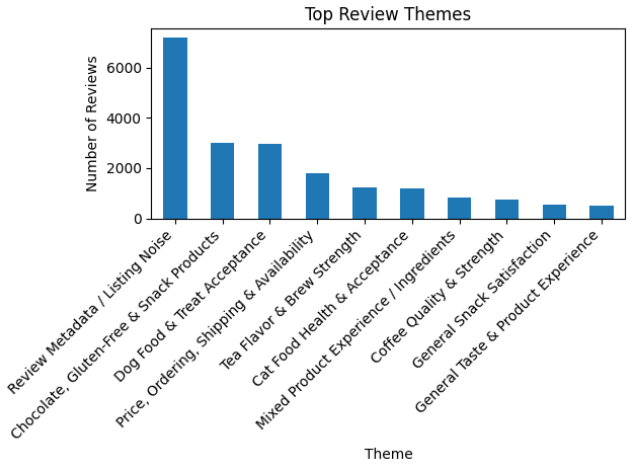


Figure 1: Theme distribution

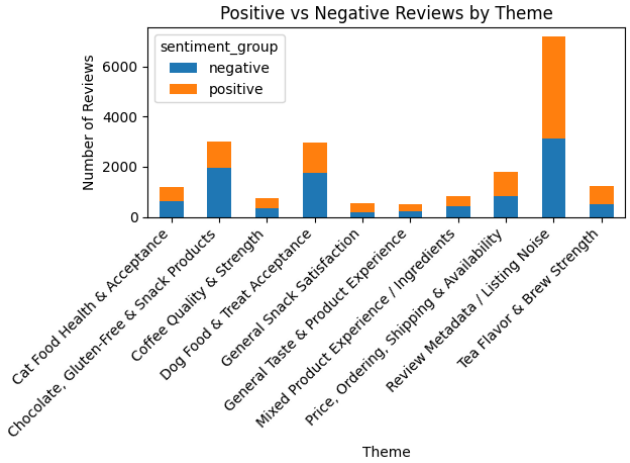


Figure 2: Sentiment by theme

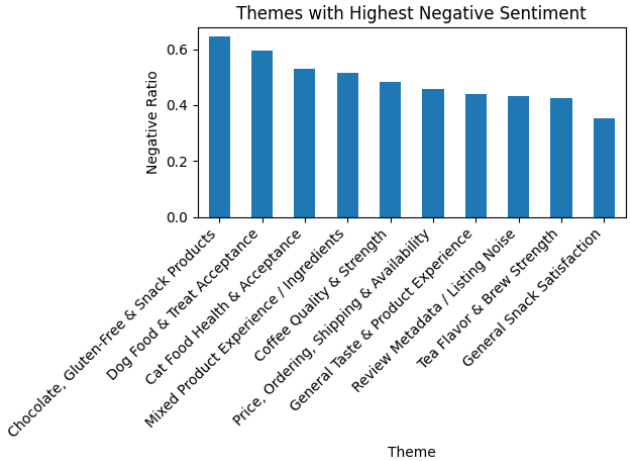


Figure 3: Highest negative sentiment themes

## Business Recommendations

| Theme | Insight | Recommendation |
|---|---|---|
| Chips & Snacks | Taste and texture drive satisfaction. | Optimize flavor profiles and highlight value positioning. |
| Tea Products | Flavor strength and consistency matter. | Improve brewing guidance and batch consistency. |
| Chocolate & Sweets | Texture and sweetness balance shape reviews. | Refine formulation and improve texture quality. |
| General Products | Value mismatches drive complaints. | Align pricing and improve product descriptions. |
| Dietary Products | Ingredient trust is critical. | Improve transparency and product quality. |
| Shipping & Orders | Logistics strongly impact sentiment. | Improve packaging and fulfillment accuracy. |
| Dog Products | Pet acceptance varies widely. | Expand palatability and safety testing. |
| Cat Products | Health outcomes drive trust. | Improve ingredient quality and health communication. |
| Coffee | Strength and roast consistency matter. | Clarify flavor profiles and improve consistency. |
| Taste Complaints | Poor flavor drives dissatisfaction. | Reformulate low-performing products. |

## Conclusion

AI-assisted clustering transformed unstructured customer reviews into actionable product, logistics, and customer-experience insights. The project demonstrates how text analytics can help companies prioritize product improvements, reduce preventable complaints, and better understand the drivers of customer satisfaction.

## Project Summary

- Dataset: 500,000+ Amazon reviews
- Modeling sample: 20,000 positive and negative reviews
- Tools: Python, pandas, TF-IDF, KMeans clustering, AI-assisted labeling
- Output: customer themes, sentiment comparisons, negative-risk analysis, and business recommendations

## Limitations and Next Steps

This analysis used a 20,000-review modeling sample from a larger 500,000+ review dataset. Because clusters were generated from review text, some themes may combine multiple related product categories. Future improvements could include product-level segmentation, time-trend analysis, and comparison against a second dataset such as electronics or app reviews.

## Technical Skills Demonstrated

Python, pandas, text cleaning, TF-IDF vectorization, KMeans clustering, sentiment segmentation, AI-assisted theme labeling, data visualization, and business recommendation development.